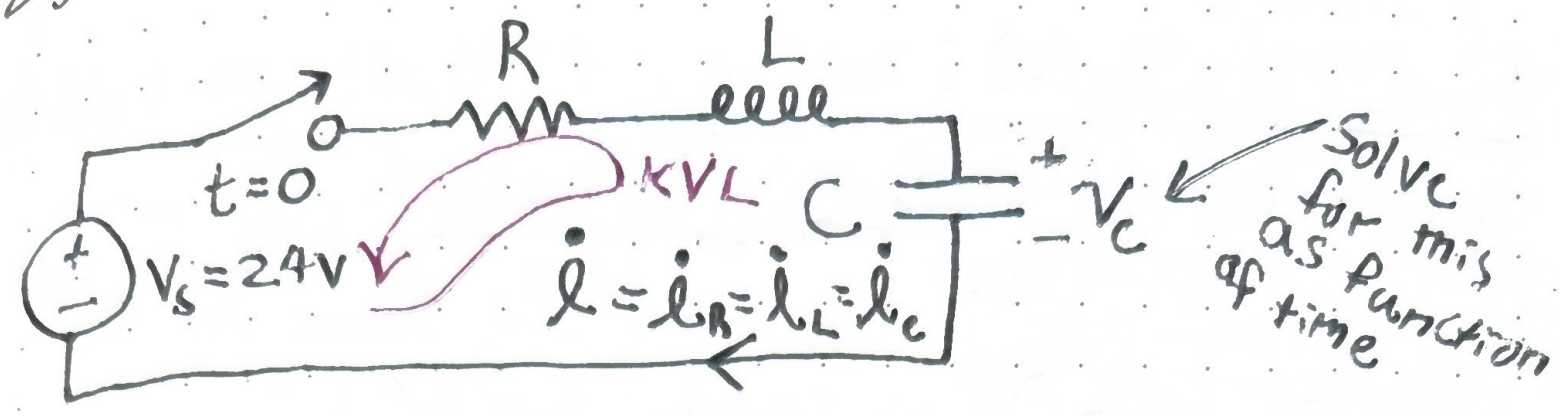
251027 2nd Order Circuits



KVL: $v_c + v_L + v_R - V_s = 0$ (for $t \geq 0$)

Rewrite v_L and v_R:

$$\Rightarrow v_L = L\frac{di}{dt} \qquad v_R = iR$$

$$\Rightarrow v_c + L\frac{di}{dt} + iR = V_s$$

Rewrite i as i_c:

$$\Rightarrow i = i_c = C\frac{dv_c}{dt} \Rightarrow L\frac{di}{dt} = LC\frac{d^2v_c}{dt^2}$$

$$\Rightarrow v_c + LC\frac{d^2v_c}{dt^2} + RC\frac{dv_c}{dt} = V_s$$

$$\boxed{\therefore \frac{d^2v_c}{dt^2} + \frac{R}{L}\frac{dv_c}{dt} + \frac{1}{LC}v_c = \frac{V_s}{LC}}$$

Form of Solution: $v_c(t) = v_{c,homo}(t) + v_{c,part}(t)$

- Solution for RHS = 0
- Since RHS (i.e., V_s/LC) does not depend on t, this is $v_c(\infty)$

$$\boxed{\therefore v_c(t) = v_{c,homo}(t) + v_c(\infty)}$$

Find $V_{c,homo}(t)$:

We have $\frac{d^2 V_{c,homo}}{dt^2} + a\frac{dV_{c,homo}}{dt} + bV_{c,homo} = 0$

where $a = R/L$ and $b = 1/LC$

→ Guess $V_{c,homo} = Ae^{st}$

Plug our guess into the diff-eq:

$\Rightarrow s^2Ae^{st} + asAe^{st} + bAe^{st} = 0$

Divide by Ae^{st}:

$\Rightarrow s^2 + as + b = 0$

Use Quadratic Formula to find s:

$\Rightarrow s = \frac{-a \pm \sqrt{a^2 - 4b}}{2}$

Thus s has 2 solutions:

$\Rightarrow s_{1,2} = -\left(\frac{a}{2}\right) \pm \sqrt{\left(\frac{a}{2}\right)^2 - b}$

$$\boxed{\therefore V_c(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + V_c(\infty)}$$

let $\alpha = a/2 \Rightarrow \boxed{\alpha = \frac{R}{2L}}$

and $\omega_0 = \sqrt{b} \Rightarrow \boxed{\omega_0 = \frac{1}{\sqrt{LC}}}$

$\Rightarrow \boxed{s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}}$

* Notice that $S_{1,2}$ can be <u>real or complex</u> depending on relationship of α and ω_0

* We need <u>two initial conditions</u> to find A_1 and A_2; assume we know <u>$v_c(0)$ and $i_c(0)$</u>

<u>Case where $S_{1,2}$ is real (overdamped):</u>

→ For $S_{1,2}$ real $\Rightarrow$ $\boxed{\alpha > \omega_0}$

Apply initial conditions:

$$\Rightarrow v_c(0) = A_1 + A_2 + v_c(\infty)$$

$$\Rightarrow i_c(0) = C\left.\frac{dv_c}{dt}\right|_{t=0} = C\left(S_1 A_1 e^{S_1 t} + S_2 A_2 e^{S_2 t}\right)\Big|_{t=0}$$

$$\Rightarrow i_c(0) = C\left(S_1 A_1 + S_2 A_2\right)$$

Solving for A_1 and A_2:

$$\Rightarrow \boxed{A_1 = \frac{i_c(0)/C - S_2\left(v_c(0) - v_c(\infty)\right)}{S_1 - S_2}}$$

$$\Rightarrow \boxed{A_2 = \frac{i_c(0)/C - S_1\left(v_c(0) - v_c(\infty)\right)}{S_2 - S_1}}$$

Case where $S_1 = S_2$ (critically damped):

→ For $S_1 = S_2 \Rightarrow \boxed{\alpha = \omega_0}$

$\Rightarrow \boxed{R = 2\sqrt{\frac{L}{C}}} \Rightarrow S_1 = S_2 = -\alpha$

Simplify solution for v_c:

$$\Rightarrow v_c(t) = A_1 e^{-\alpha t} + A_2 e^{-\alpha t} + v_c(\infty)$$

$$\Rightarrow v_c(t) = (A_1 + A_2) e^{-\alpha t} + v_c(\infty) = A_3 e^{-\alpha t} + v_c(\infty)$$

- But A_3 is only one unknown, and we need to apply two initial conditions
- Modify the form w/ a new guess:

$$v_c(t) = B_1 e^{-\alpha t} + B_2 t e^{-\alpha t} + v_c(\infty)$$

$$\boxed{\therefore v_c(t) = (B_1 + B_2 t) e^{-\alpha t} + v_c(\infty)}$$

Apply initial conditions:

$$\Rightarrow v_c(0) = B_1 + v_c(\infty)$$

$$\Rightarrow i_c(0) = C \left.\frac{dv_c}{dt}\right|_{t=0} \Rightarrow i_c(0) = C(-\alpha B_1 + B_2)$$

Solving for B_1 and B_2:

$$\Rightarrow \boxed{B_1 = v_c(0) - v_c(\infty)}$$

$$\Rightarrow \boxed{B_2 = \frac{i_c(0)}{C} + \alpha\left(v_c(0) - v_c(\infty)\right)}$$

Case where $S_{1,2}$ is complex (under damped):

→ For $S_{1,2}$ complex ⇒ $\boxed{\alpha < \omega_0}$

let $\omega_d^2 = \omega_0^2 - \alpha^2$

$$\Rightarrow S_{1,2} = -\alpha \pm \sqrt{-\omega_d^2}$$

$$\Rightarrow \boxed{S_{1,2} = -\alpha \pm \omega_d j \quad \text{where } j = \sqrt{-1}}$$

Rewrite solution for v_c:

$$\Rightarrow v_c(t) = A_1 e^{-\alpha t} e^{j\omega_d t} + A_2 e^{-\alpha t} e^{-j\omega_d t} + v_c(\infty)$$

Use Euler's Identity ($e^{\pm j\theta} = \cos(\theta) \pm j\sin(\theta)$) and replace $e^{\pm j\omega_d t}$:

$$\Rightarrow v_c(t) = A_1 e^{-\alpha t}\left(\cos(\omega_d t) + j\sin(\omega_d t)\right)$$
$$+ A_2 e^{-\alpha t}\left(\cos(\omega_d t) - j\sin(\omega_d t)\right) + v_c(\infty)$$

let $D_1 = A_1 + A_2$ and $D_2 = j(A_1 - A_2)$

$$\boxed{\therefore v_c(t) = e^{-\alpha t}\left(D_1 \cos(\omega_d t) + D_2 \sin(\omega_d t)\right) + v_c(\infty)}$$

After applying initial conditions $v_c(0)$ and $i_c(0)$, we will find:

$$\boxed{D_1 = v_c(0) - v_c(\infty)}$$

$$\boxed{D_2 = \frac{i_c(0)/C + \alpha\left(v_c(0) - v_c(\infty)\right)}{\omega_d}}$$

falstad ex/

Set $C = 720\ \mu F$

and $L = 300\ mH$

=> For critically damped

$$R = 2\sqrt{\frac{L}{C}} = 2\sqrt{\frac{300 \times 10^{-3}}{720 \times 10^{-6}}} = \underline{\underline{41\ \Omega}}$$